PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – I

INTRODUCTION

Syllabus : Principles of Pre-Stressing – Prestressing Systems – Pre-Tensioning and Post-Tensioning – Advantages and Limitations of Pre-Stressed Concrete – Need for High Strength Materials. Methods of Pre-Stressing : Pre-Tensioning (Hoyer System) and Post-Tensioning (Freyssinet System and Gifford-Udall System).

Contents of this unit

1. Introduction & basic concept of prestressing
2. Principles of prestressing (3 concepts)
3. Types / classification of prestressing
4. Pre-tensioning – procedure
5. Post-tensioning – procedure
6. Pre-tensioning vs Post-tensioning
7. Advantages, limitations & applications
8. PSC vs RCC
9. Need for high strength concrete & steel
10. Hoyer system
11. Freyssinet system
12. Gifford-Udall system (+ Magnel-Blaton, Lee-McCall)
13. Important questions & quick revision

Exam tip : Draw neat sketches for every system (Hoyer, Freyssinet, Gifford-Udall) – 50 % of marks are for the diagram and the sequence of operations.

1. Introduction

Concrete is strong in compression (f_{ck} high) but very weak in tension (≈ 10 % of f_{ck}). So plain concrete beams crack early under load.

- In RCC, steel takes the tension **after** concrete has cracked → cracks are always present, concrete below N.A. is wasted.
- In PSC, a **compressive force is applied in advance** (before external loads) so that tension is cancelled / reduced.

Definition (T.Y. Lin) : Prestressed concrete is concrete in which there have been introduced internal stresses of such magnitude and distribution that the stresses resulting from the given external loadings are counteracted to a desired degree.

Basic idea – the "stack of books" example

Pick up a row of books by pressing them horizontally at the ends – the row acts like a beam and carries its own weight even though there is no tensile strength between books. The squeezing (prestress) makes the compressive force; friction holds them. Similarly a rope of bricks / barrel with iron hoops (circular prestress).



books squeezed by horizontal force P act as a beam

Brief history

- 1886 – P.H. Jackson (USA) – first patent for tightening steel rods in concrete arches.
- 1888 – Doehring (Germany) – prestressed slabs with metal wires. All early attempts **failed** – low-strength steel, losses wiped out the prestress.
- 1928 – **Eugène Freyssinet** (France) – used **high-strength steel** and high-grade concrete; the father of PSC. Shrinkage/creep losses understood.
- 1940s – Magnel (Belgium), Hoyer (Germany) – practical systems. 1950s onwards – bridges, tanks, poles, sleepers.
- India – first PSC bridge near Palghat (Kerala) 1948; IS 1343 is the code of practice (latest 2012).

Key point : Prestress must be of high magnitude, so **high strength steel is a must** – because losses in ordinary steel ≈ prestress itself.

2. Principles of Pre-stressing

Three ways of looking at prestressed concrete (Lin / Guyon):

(i) Prestressing to transform concrete into an elastic material (Stress concept – Freyssinet)

- Concrete = brittle material, poor in tension. Prestressing converts it into an **elastic, un-cracked** material.
- Compressive prestress is applied so that net tension under working loads is **zero or within the permissible limit**.
- Concrete section is analysed as a homogeneous elastic material – ordinary bending theory applies.

$$\text{Concentric : } f = P/A \pm M \cdot y/I$$

$$\text{Eccentric : } f = P/A \pm P \cdot e \cdot y/I \pm M \cdot y/I$$

where P = prestressing force, A = area of section, e = eccentricity, I = moment of inertia, y = distance from C.G., M = external bending moment. (Compression +ve, tension –ve in this note.)

(ii) Prestressing for combination of high-strength steel with concrete (Strength concept)

- PSC is a **combination** of high strength steel and concrete – like RCC.
- Steel takes tension T, concrete takes compression C, forming an **internal resisting couple** of lever arm z.
- Difference from RCC : steel is pre-tensioned, so the whole concrete section is active and the lever arm is larger and almost constant before / after cracking.

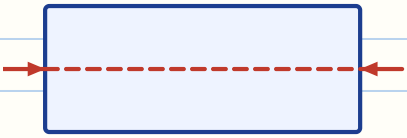
$$M_u = T \times z = C \times z$$

(iii) Prestressing to achieve load balancing (T.Y. Lin)

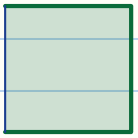
- Prestressing is an attempt to **balance** a part of the external load by the transverse (upward) force from the curved tendon.
- If the net load is balanced, beam has only axial compression (uniform stress) → no deflection, no cracking.

Stress distribution diagrams

(a) Concentric prestress



P through C.G.



uniform P/A

(b) Flexure by loads



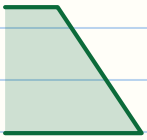
M·y/I : compression top, tension bottom

Resultant : $f = P/A \pm M \cdot y/I$ (zero tension if $P/A \geq M \cdot y/I$)

(c) Eccentric prestress



P at e below C.G.



trapezoid

Self-weight stress + prestress



top $P/A - Pe/Z + M/Z$

bottom $P/A + Pe/Z - M/Z$

Eccentric prestress produces hogging (upward camber) – opposes sagging due to loads.

Camber $\approx P \cdot e \cdot L^2 / (8\ E\ I)$ for parabolic cable with constant P

Load balancing : Parabolic tendon with dip e in span L, force P exerts an **upward UDL** = $8 \cdot P \cdot e / L^2$. For a balanced load w_b:

$$w_b = 8\,P\,e / L^2 \rightarrow P = w_b L^2 / (8\,e)$$

Worked example (load balancing)

A beam of span 10 m carries w = 20 kN/m (total). Tendon parabolic with e = 0.25 m. Find P to balance the load completely.

$$P = w L^2 / (8e) = 20 \times 10^2 / (8 \times 0.25) = 2000/2 = 1000\ kN$$

Under this load the beam is under uniform compression P/A only – no bending, no deflection.

3. Types / Classification of Prestressing

Basis	Types	Remarks
Degree of prestress	**Full** – no tension at all under working load (Type 1) **Limited** – tension within permissible, no cracking (Type 2) **Partial** – cracks allowed but width limited (Type 3)	Type 3 needs additional mild steel (hybrid)
Method of tensioning	**Pre-tensioning** – steel tensioned before concreting **Post-tensioning** – after concrete has hardened	Both are covered in this unit
Bond	**Bonded** – grouted ducts / direct bond **Un-bonded** – greased & sheathed strands	Unbonded: slabs, buildings
Location of tendon	**Internal** – inside concrete **External** – outside the section (box girder, strengthening)	External can be re-stressed
Shape of member	**Linear** – beams, slabs, girders **Circular** – tanks, pipes, silos, domes (wire-wound)	
Tendon profile	Straight, bent (harped), parabolic, circular	
Source of stress	Mechanical (jacks), hydraulic, electrical (heating), chemical (expanding cement)	Hydraulic jacks most common

Prestressing systems – overview

- **Pre-tensioning :** Hoyer system (long line bed, bond transfers stress).
- **Post-tensioning :** Freyssinet (cone & wedge), Gifford-Udall (single wire, cone anchorage), Magnel-Blaton (sandwich plates, wedges), Lee-McCall (threaded bars), CCL, VSL, BBRV (button heads).
- Classification on anchoring : wedge action (Freyssinet, Gifford-Udall, Magnel), direct bearing (button-head BBRV, nuts & threaded bar – Lee-McCall), looped (Baur-Leonhardt, Leoba).

Stages in any system : (1) tensioning → (2) anchoring → (3) transfer of prestress to concrete. Remember the three words **Tension – Anchor – Transfer**.

4. Pre-Tensioning

Steel wires/strands are **tensioned first**, concrete is cast around them, and when concrete attains sufficient strength the tendons are released – stress is transferred to concrete by **bond** (and transmission length).

Procedure (step by step)

1. Long casting bed (50–100 m) with **anchor abutments** at both ends.
2. Tendons are laid along the bed and **tensioned by hydraulic jack** against the abutment; anchored with grips / chucks.
3. Moulds are placed, reinforcement cages (stirrups) fixed, and **concrete is poured** and compacted – tendons remain stretched.
4. Concrete is cured (steam curing speeds up) until it attains **transfer strength ≥ 0.5 – 0.7 f_{ck}** (≈ 3–4 days with steam).
5. Tendons are **gradually released** (jacks / sand boxes / cutting by flame – gradual). Steel tries to shorten → grips the concrete → compression in concrete.

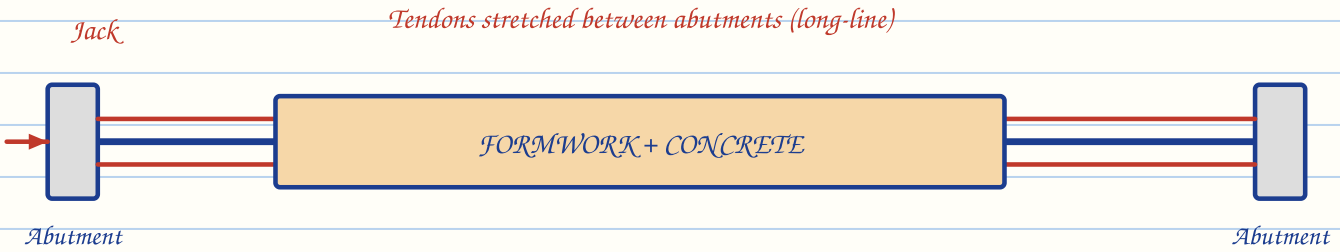


Fig. Pre-tensioning – long line method

Key points : No anchorage device in the member (bond is the anchor) • Tendons are generally straight or harped • Suitable for **factory production** of small / medium precast units – poles, sleepers, planks, beams, pipes • Losses : elastic shortening, shrinkage, creep, relaxation (no friction, no anchorage slip in member).

Transmission length

Length from the end over which the prestress builds up fully by bond. Depends on wire diameter, strength of concrete, surface (indented wires better). IS 1343 : $L_t \approx 30–50\ \varphi$ ... (detailed in Unit III).

5. Post-Tensioning

Concrete is cast first (with **ducts / sheaths** left in it); when it has gained strength the tendons are **stretched against the hardened concrete** using jacks and anchored at the ends. Stress is transferred by **bearing** through end anchorages.

Procedure (step by step)

1. Cast concrete member with a **duct** (corrugated metal / HDPE sheath) placed to the required profile (straight / curved) – with end **anchor blocks** and helix / bursting steel.
2. Cure until concrete reaches the specified **transfer strength** ($\geq 0.7\, f_{ck}$ or as per design, min 25–30 MPa).
3. Thread tendons (wires / strands / bars) through the ducts.
4. **Stress** the tendons with hydraulic jack reacting against the member itself; lock the **anchorage** (wedge / nut / button head). Jack force is held while anchoring.
5. **Grout** the ducts (cement + water + admixture, w/c ≈ 0.40–0.45) – protects steel from corrosion and gives bond (bonded post-tensioning).

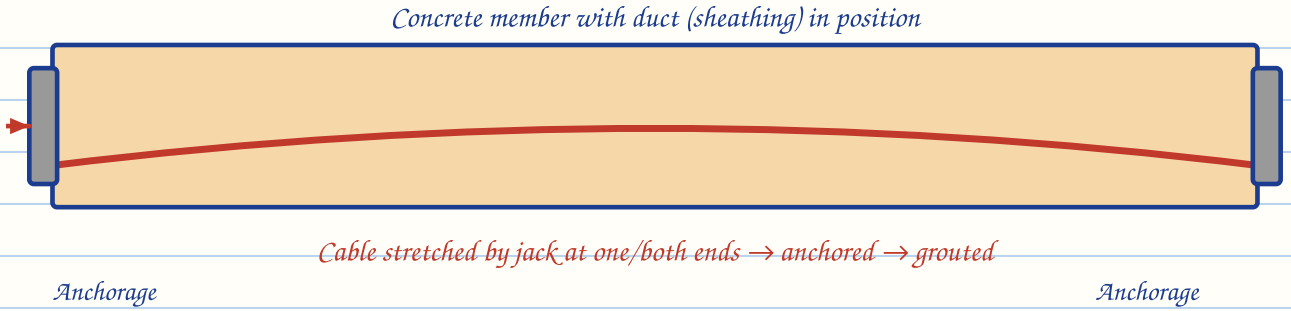


Fig. Post-tensioning schematic

Key points : Anchorage is essential – transfers P by bearing • Curved (parabolic) tendons possible → ideal for long-span bridges, large girders • Done at site (cast in-situ) or on precast units • Losses : elastic shortening (average), shrinkage, creep, relaxation, **friction** and **anchorage slip**.

Remember : Pre-tension = "Tension, then cast". Post-tension = "Cast, then tension".

6. Pre-tensioning vs Post-tensioning

Sl.	Pre-tensioning	Post-tensioning
1	Tendons stretched **before** casting concrete	Tendons stretched **after** concrete hardens
2	Stress transferred by **bond**	Stress transferred by **bearing** (anchorage)
3	No anchorage hardware in member – cheaper per member	Anchorages required – costly
4	Needs long casting beds, abutments – heavy initial investment	Can be done on site; no abutment needed
5	Suitable for small/medium precast mass-produced units	Suitable for large members, long spans, cast-in-situ
6	Straight or harped tendons only (bent at deviators)	Any profile – parabolic, curved easily
7	Eccentricity limited; profile control difficult	Profile controlled by ducts
8	Transmission length at the ends needed	Anchorage zone – end block, bursting steel
9	Losses : elastic shortening (full), creep, shrinkage, relaxation	Also friction + anchorage slip; elastic shortening partly nil if tendons stressed one by one → small
10	Grouting not needed	Grouting of ducts needed (for bonded)
11	Quick production, steam curing, high economy for repetition	Slower, labour intensive
12	Examples : railway sleepers, poles, hollow-core slabs, T-beams	Examples : bridge girders, tanks, containment, long beams

Frequently asked : **"Distinguish between pre-tensioning and post-tensioning" – 5 marks**. Write at least 8 points as above.

7. Advantages of Prestressed Concrete

1. **Section is fully utilised** – whole depth is un-cracked, so I_{gross} is used. Lesser depth, lighter members.
2. **Saving in materials** – 20–35 % concrete and 60–70 % steel (HTS), though steel is costlier per kg.
3. **No cracking** under working loads – better durability, corrosion protection, water tightness (tanks).
4. **Reduced deflection** – camber counteracts load deflection; slender members and long spans (up to 100 m +).
5. **Improved shear resistance** – principal tension reduced by axial compression; web can be thinner.
6. **Better fatigue & impact resistance** – stress range in steel is small.
7. **High-quality control** – the member is tested in effect during stressing (a "proof load").
8. **Quick erection** – precast components, early removal of formwork; less maintenance.
9. **Elastic recovery** – cracks (if any under overload) close on removal of overload.
10. Used for **special structures** – bridges, nuclear containment, tanks, offshore platforms, water towers.

Limitations / Disadvantages

1. Needs **high-quality materials** (HSC & HTS) – costlier, availability.
2. **Skilled labour** and special equipment (jacks, anchors, grouting) required.
3. **Initial cost** (casting bed, anchorage hardware) high – not economical for small spans / low loads.
4. Prestress **losses** must be estimated carefully – loss of effectiveness over time.
5. Needs **greater supervision** and quality control at every stage – a mistake can be disastrous.
6. **Fire resistance** is lower – HTS loses strength above 400 °C; more concrete cover needed.
7. Brittle failure risk if over-prestressed (over-stress / ultimate behaviour).
8. Difficulty in **repair, strengthening, demolition** (stored energy).
9. Anchorage zone needs **special reinforcement** (bursting, spalling).

Applications

Bridges (girders, box, cable-stayed), industrial floors, roofs, silos, tanks, pipes, poles, railway sleepers, piles, dams (anchors), nuclear reactor vessels, offshore towers, runway pavement, retaining walls & rock anchors.

8. Prestressed Concrete vs Reinforced Concrete

Aspect	RCC	PSC
Section	Cracked below N.A.; only part (compression zone) effective	Whole section effective, un-cracked
Steel	Mild steel / HYSD, f_y = 415–550 MPa; stress ≈ 140–230 MPa	HTS, f_{pk} = 1570–1860 MPa; working stress up to ≈ 1000+ MPa
Concrete	M20 – M30 typical	M35 – M60 (higher)
Deflection	Large; creep deflection more	Small (camber)
Span / depth	L/d = 20–26	L/d = 25–35, longer spans
Cracks	Always present at service	Not present (full prestress)
Self-weight	Heavy, deeper sections	Light, slender sections
Stress state	Steel stress varies with loads	Steel stress nearly constant
Shear	Needs more stirrups	Reduced by prestress
Skill/Cost	Easy; low cost for short span	Skilled; economical for long span

9. Need for High Strength Materials

(a) High strength concrete

- **High compressive strength** is needed to resist the high stresses at transfer – stress concentration near anchorages (bearing stresses).
- Smaller creep & shrinkage strain (higher E_c) → **smaller losses**.
- Higher **bond strength** (pre-tension) and **shear** strength.
- Higher tensile strength → resists cracking (modulus of rupture $f_{cr} = 0.7\sqrt{f_{ck}}$).
- Economy – smaller sections, lighter members, faster (early) strength gain permitting early transfer.
- IS 1343 : min M30 for post-tensioned, M40 for pre-tensioned members.

(b) High strength steel – why mild steel will not do

Consider a 6 m long member. Initial prestress in the steel must be large because **losses are fixed strain losses** (shrinkage + creep + elastic ≈ 0.0007–0.0015 m/m in total).

Example : Total loss strain in concrete (shrinkage + creep + elastic) ≈ **0.0008** (typical). Steel $E_s = 2 \times 10^5$ N/mm².

$$\text{Loss of stress} = \varepsilon \times E_s = 0.0008 \times 2\times10^5 = 160 \text{ N/mm}^2$$

Steel	Initial stress	Loss	Remaining	% loss
Mild steel	140 N/mm²	160	**–20 (nil!)**	> 100 %
HTS wire	1200 N/mm²	160 (+relax.)	1040 N/mm²	≈ 13 %

So with mild steel the prestress is completely lost; with HTS the effective prestress is retained.

Hence high initial stress → high-strength steel.

Requirements of prestressing steel

- High tensile strength (1570–2000 N/mm²) and high yield (proof stress 0.2 % ≥ 0.85 f_{pk}).
- Adequate **ductility** (min. elongation ≥ 3.5 % on gauge length 200 mm).
- Low **relaxation** (low-relaxation strands ≈ 2.5 % at 1000 h).
- Good bond / surface (indented, crimped, stranded), freedom from corrosion, and ability to be anchored.
- Straightness – coiled wire diameter large enough to avoid residual curvature.

Types of prestressing steel (IS 1785, IS 6006, IS 2090)

Type	Details	f_{pk} (N/mm²)
Cold-drawn wires (IS 1785)	Ø 2.5, 3, 4, 5, 7, 8 mm; plain or indented	1570–1860 (smaller dia → higher)
Strands (IS 6006)	7-wire standard / super; 2, 3 wires also; Ø 9.5, 12.7, 15.2 mm	1770–1860
High-tensile alloy bars (IS 2090)	Ø 10–32 mm, threaded; used with nuts (Lee-McCall)	980–1030

Remember : Wire Ø 7 mm → f_{pk} = 1570 MPa; strand 12.7 mm → 1860 MPa (IS 6006). Modulus $E_s = 2.0 \times 10^5$ N/mm² (wires, bars) and 1.95×10^5 (strands).

10. Methods of Pre-stressing

A. Pre-tensioning : HOYER SYSTEM (Long-line system)

Developed by **E. Hoyer (Germany)** – also called the Hoyer long-line system / "Hoyer effect".

- Piano wires (Ø 2–8 mm) are tensioned between two **massive abutments** 100 m or more apart on a **long casting bed**.
- Many members (e.g. 10–20 sleepers, joists) are cast in line on the same wires, separated by **bulkheads / separators**.
- After concrete has hardened, the wires are **released gradually** by **jacks / sand jacks** at the tension end – so wire tries to regain length, swells in diameter at the cut end (Hoyer effect) and gives a wedge-action bond.
- The wires are then **cut between the members** – each member is separated.

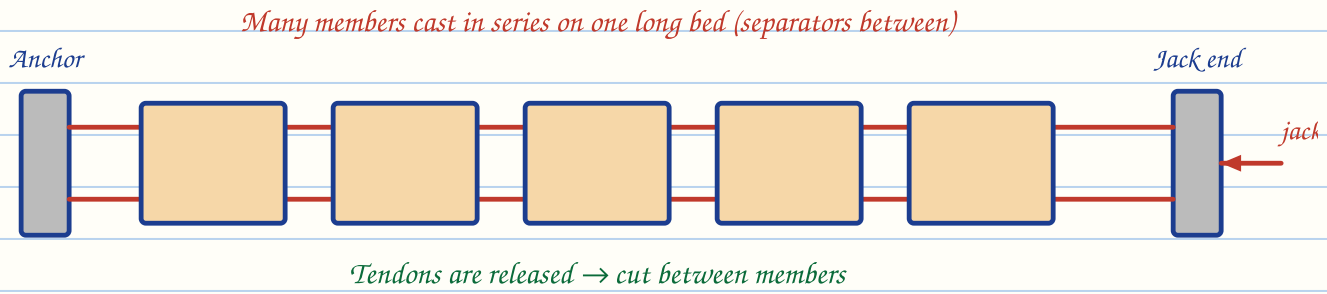


Fig. Hoyer long-line pre-tensioning system

Hoyer effect

When tension is released the wire contracts laterally (Poisson effect) except inside the concrete – the diameter increases back inside the member, producing **wedge action** that improves the bond at the ends and aids stress transfer.

Advantages : economical for mass production, no end anchorage, quick re-use of bed, steam curing.

Limitations : long bed, loss in length of wire between members (waste), only straight tendons, release must be gradual to avoid shock.

B. Post-tensioning : FREYSSINET SYSTEM

Developed by **Eugène Freyssinet (France)**. Wire / strand tendons with **cone and wedge anchorage**.

- Tendon = **12 wires Ø 5 mm or Ø 7 mm** (or 12 strands of 12.7 / 15.2 mm in later versions) placed in a flexible metal duct (sheath).
- Anchorage : a **concrete (or steel) cylindrical female cone** embedded in the end of the member + a **grooved male (wedge) cone** – wires lie in the grooves.
- Jacking : a **double-acting hydraulic jack** pulls the wires; after the required force is reached, the male cone is **forced (pressed) into the female cone** by the 2nd (inner) ram – wedge action grips wires.
- Jack released → wires try to shorten but the wedge grips → prestress transferred by bearing at the anchor.
- Ducts **grouted** under pressure (cement grout) – bond + corrosion protection.

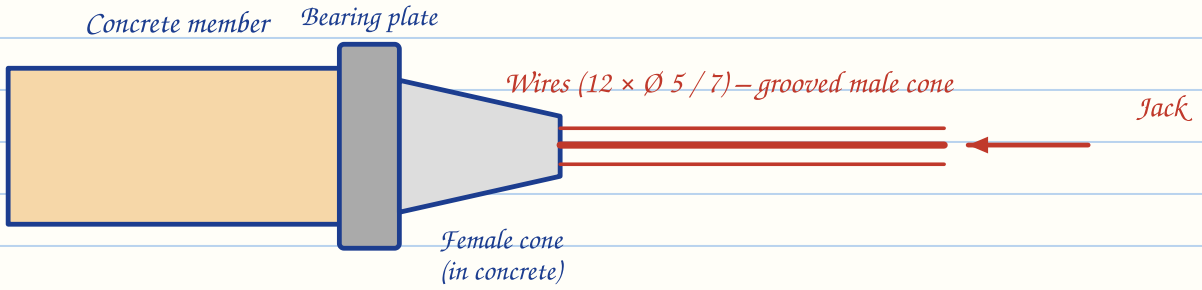


Fig. Freyssinet cone anchorage – principle

Salient features

- Cone angle small (≈ 12°) → good wedge grip. Duct ≈ 28–40 mm Ø. Each tendon takes ≈ 400–1000 kN.
- Wedge-type, **tension at one or both ends**; many tendons can be placed in a section – used for bridges, tanks, silos.
- Anchor slip (wedge draw-in) ≈ 6 mm – to be accounted in losses (Unit II).
- Dead end anchorage, "Freyssinet jack" is of 2 stages – pulling and wedging.

Sequence : Cast – Insert tendons – Fit jack – Stress – Push wedge – Release – Grout.

C. Post-tensioning : GIFFORD-UDALL SYSTEM

British system (Gifford & Udall, 1939) – a **single-wire system** using a very simple **cone grip** anchorage; the **Gifford-Udall-CCL** variant is widely used.

- Each wire (Ø 2–8 mm, usually 5–7 mm) is **stressed individually** by a small **single-wire jack** (CCL jack).
- The wire is passed through an **anchor plate (bearing)** and a split **conical wedge (grip)** – wedges are **pushed in by the jack** after stretching.
- Wires are placed in ducts in groups; in strand version (CCL) 7-wire strands with split-barrel anchors are used.
- After stressing, **wedges lock** the wire; excess length cut off; ducts grouted.

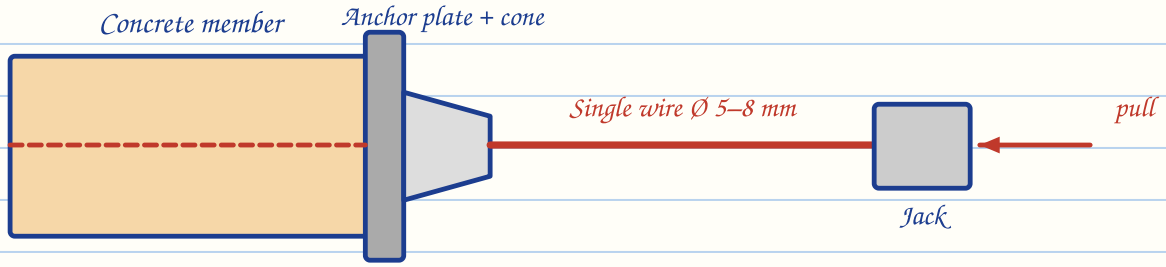


Fig. Gifford-Udall anchorage – cone grip

Features

- Simple equipment, easily portable – used for **small / medium** members as well as large.
- Wires stressed **one by one** → elastic shortening loss is smaller (pre-stress in the first wires is reduced by later wires but can be compensated).
- Anchorage is cheap; loss of prestress by **slip ≈ 1–2 mm**(wedge draw-in) to be allowed for.
- Not suitable if all wires need to be stressed simultaneously (large tendons – use Freyssinet).

Other post-tensioning systems (short notes)

- **Magnel-Blaton (Belgium)** – flat sandwich plates with 2 wires per wedge; 4-wires per layer, 8 wires/tendon; uses steel wedges between sandwich plates – two wires are stressed simultaneously.
- **Lee-McCall** – high-tensile alloy steel **bars** with threaded ends and **nuts** – direct bearing anchorage; low loss (no slip).
- **BBRV** – button-heads formed on wires, bearing on the anchor plate. **VSL** – strands with wedge anchor.

11. Important Questions & Quick Revision

Short answers (2 marks)

1. Define prestressed concrete.
2. State the three concepts of prestressing.
3. Why is high strength steel necessary for prestressing?
4. What is the Hoyer effect?
5. List the differences between bonded and un-bonded tendons.
6. What is transmission length?
7. Name any two post-tensioning systems.

Long answers (10 marks)

1. Explain the basic principles of prestressing with neat sketches. Compare it with RCC.
2. Distinguish pre- and post-tensioning. State the merits and demerits of PSC.
3. Explain the Hoyer system of pre-tensioning with a neat sketch.
4. Describe the Freyssinet system of post-tensioning.
5. Explain the Gifford-Udall system. How does it differ from Freyssinet system?
6. Why do we need high-grade concrete and HTS in prestressing? Give an example.

Quick revision formulae & facts

Item	Value / formula
Stress at fibre (eccentric P)	$f = P/A \pm P\cdot e\cdot y/I \pm M\cdot y/I$
Load balancing (parabolic)	$w_b = 8Pe / L^2$
Min grade (IS 1343)	Post-tension M30, pre-tension M40
Loss strain (typical)	$\varepsilon \approx 0.0008 \rightarrow 160\ N/mm^2$ for $E_s = 2\times10^5$
Transfer strength	$\geq 0.5–0.7\ f_{ck}$
Stage sequence	Tension → Anchor → Transfer
Pre-tension	Bond; Hoyer; long-line
Post-tension	Bearing; Freyssinet (12 wires cone), Gifford-Udall (single wire jack), Magnel (sandwich), Lee-McCall (bars)

Best of luck ! Practise the diagrams – they carry the marks.